A Provable Framework for Measuring and Mitigating Subgroup Representation Bias with Proxy Attributes

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Abstract

When an AI system performs worse for certain demographic groups, how do you detect and fix the disparity if you cannot observe who belongs to which group? Protected attribute labels (race, sex, skin type) are often unavailable due to privacy, legality, or collection cost. This paper provides an answer: use a *proxy variable*—any observable measurement correlated with the protected attribute—to rebalance the training data. We prove a theorem bounding the worst-case accuracy disparity (SBAG) after proxy-stratified training, showing it decomposes into a generalization term, a proxy information gap $(1 - \rho)\Phi$, and a confidence term. We validate across medical imaging (melanoma detection), tabular income prediction, and real text (20 Newsgroups). In a controlled medical experiment, our method significantly improves accuracy for *all* subgroups ($p < 0.001$). When the proxy is adequate ($\rho = 0.23$, sex via hours-per-week), it achieves a statistically significant 27% SBAG reduction ($p < 0.0001$). Four failure modes and a negative control characterize the method's boundaries. Code is provided for full reproducibility.

1 Introduction

Consider a dermatology AI system used to screen for melanoma. If it performs accurately on light skin but misses melanomas on dark skin, patients with darker skin tones are harmed. Fixing this requires knowing each patient's skin type—but dermatologists rarely record this, hospitals may not collect it, and patients may not know it. This is the *fairness under unawareness* problem: how do you measure and mitigate performance disparities across subgroups when you cannot observe subgroup membership?

This paper provides an answer. We introduce **P-Stratified Sampling**, a theoretically grounded method that uses any observable *proxy variable* correlated with the unobserved protected attribute to rebalance training data. The core idea is simple: bin the training data by the proxy value, then oversample each bin to equal size. This forces the model to train on a distribution that is balanced across the proxy range, which—by correlation—balances across the true subgroups.

Our main theoretical contribution is a finite-sample bound on the Subgroup Balanced Accuracy Gap (SBAG) after P-Stratified training. The bound has three interpretable terms:

- **Generalization** ($\propto \sqrt{m/n}$): the price of splitting data into m bins.
- **Proxy gap** ($\propto 1 - \rho$): the cost of an imperfect proxy, where ρ is the proxy-attribute correlation.
- **Confidence** ($\propto \sqrt{\log(1/\delta)/n}$): statistical uncertainty.

When the proxy is perfect ($\rho = 1$), the bound reduces to the standard per-subgroup guarantee. When the proxy is useless ($\rho \approx 0$), it recovers the ERM bound up to constants. The intermediate cases are where the theory guides practice.

We validate across four experiments spanning three domains—medical imaging, tabular data (two analyses), and real text—comparing P-Stratified against ERM, class-stratified learning, and proxy reweighting. Across all experiments, P-Stratified matches or outperforms alternatives. Two results reach statistical significance: a substantial accuracy improvement in medical imaging ($p < 0.001$) and a 27% SBAG reduction in the sex-based tabular analysis ($p < 0.0001$). An ablation, negative control, and four failure mode analyses provide an honest characterization of when the method helps and when it does not.

What makes this different from prior work. Existing fairness methods require protected attribute labels at training time, test time, or both. Methods for fairness "under unawareness" either estimate disparities without correction or provide heuristic mitigation without formal guarantees. Our framework is the first to simultaneously: (1) operate with only a proxy, (2) provide a finite-sample fairness bound, (3) validate across multiple domains, and (4) release fully reproducible code.

2 Problem Formulation

Consider binary classification with input space $\mathcal{X}$ and labels $\mathcal{Y} = \{0, 1\}$. Let $A \in \{0, 1\}$ be a protected attribute partitioning the population into two subgroups. Let $P \in [0, 1]$ be a proxy variable observable for every sample and correlated with A.

We receive n i.i.d. training samples $S = \{(x_i, y_i, p_i)\}_{i=1}^{n}$, where p_i is observed but a_i is not. A held-out test set has A available for evaluation.

For a classifier $h : \mathcal{X} \to [0, 1]$ and threshold $\tau \in [0, 1]$, the predicted label is $\hat{y} = \mathbf{1}[h(x) \geq \tau]$.

Definition 2.1 (Balanced Accuracy)**.** *For subgroup g, $bal_acc_g(h) = \frac{1}{2}(TP_g/(TP_g + FN_g) + TN_g/(TN_g + FP_g))$.*

Definition 2.2 (Subgroup Balanced Accuracy Gap)**.**

$$SBAG(h) = \max_{i,j} |bal_acc_{g_i}(h) - bal_acc_{g_j}(h)|.$$

Assumption 2.3 (Proxy Relevance)**.** *P and A satisfy $\rho = |corr(P, A)| > 0$. For $i \neq j$, $D_{KS}(P|A = g_i, P|A = g_j) \geq \Delta > 0$.*

3 Key Intuition

The central idea is straightforward. Suppose you have a training dataset where one subgroup is underrepresented. Standard stratified sampling would fix this by oversampling the minority subgroup—but you cannot, because you do not know who belongs to it.

If you have a proxy P correlated with the subgroup A, then bins of P contain different proportions of A samples. By *balancing the bins* (making each bin have the same number of training samples), you pull the training distribution closer to the population distribution for each subgroup. The stronger the correlation ρ, the more the proxy bins separate the subgroups, and the more effective the correction.

Algorithm 1 formalizes this: compute m quantile bins of P, count samples per bin, and oversample each bin to match the largest bin. The resulting training set S' has equal representation across the proxy range. Lemma 4.1 proves that this implies more equal representation across the true subgroups, with the benefit proportional to ρ^2.

Algorithm 1 P-Stratified Sampling

Require: $S = \{(x_i, y_i, p_i)\}_{i=1}^n$, bins m
Ensure: Balanced set S'
1: Compute quantile boundaries $q_0, \ldots, q_m$ from $\{p_i\}$
2: Assign sample i to bin b if $q_{b-1} \le p_i < q_b$
3: $n_{\max} \leftarrow \max_b |S_b|$
4: **for** $b = 1$ to m **do**
5: $\quad r_b \leftarrow \lceil n_{\max}/|S_b| \rceil$, $S'_b \leftarrow \text{repeat}(S_b, r_b)$
6: **end for**
7: $S' \leftarrow \bigcup_b S'_b$, shuffle, **return** S'

4 Theory

4.1 Lemma: How Much Does the Proxy Reveal?

The empirical Rademacher complexity of $\mathcal{H}$ on S is

$$\hat{\mathfrak{R}}_S(\mathcal{H}) = \mathbb{E}_\sigma \left[\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \sigma_i h(x_i) \right],$$

with σ_i independent Rademacher variables. The population value is $\mathfrak{R}_n(\mathcal{H}) = \mathbb{E}_{S \sim \mathcal{D}^n}[\hat{\mathfrak{R}}_S(\mathcal{H})]$.

For the balanced 0-1 loss ℓ_{bal} (which is L-Lipschitz with $L \le 1/[2\min\{\mathbb{P}(y=0), \mathbb{P}(y=1)\}]$), the standard bound [Bartlett and Mendelson, 2002] gives: with probability $\ge 1 - \delta$,

$$\mathbb{E}[\ell_{\text{bal}}(h(x), y)] \le \hat{\mathbb{E}}[\ell_{\text{bal}}(h(x), y)] + 2L \cdot \mathfrak{R}_n(\mathcal{H}) + \sqrt{\frac{\log(1/\delta)}{2n}}.$$

4.2 Key Lemma

Lemma 4.1 (Proxy Information Bound)**.** *Let A be binary with $\pi = \mathbb{P}(A = 1)$ and let $\rho = |corr(P, A)|$. Define $q(p) = \mathbb{E}[A \mid P = p]$. Then:*

$$\rho^2 \le \frac{Var(q(P))}{Var(A)} \le 1.$$

The ratio $\eta^2 = Var(q(P))/Var(A)$ is the correlation ratio (fraction of variance of A explained by P).

Proof. By the law of total covariance, $\text{Cov}(A, P) = \text{Cov}(\mathbb{E}[A \mid P], P) = \text{Cov}(q(P), P)$. Applying Cauchy-Schwarz:

$$\text{Cov}(A, P)^2 \le \text{Var}(q(P)) \cdot \text{Var}(P).$$

Substituting $\text{Cov}(A, P)^2 = \rho^2 \cdot \text{Var}(A) \cdot \text{Var}(P)$ gives $\rho^2 \cdot \text{Var}(A) \cdot \text{Var}(P) \le \text{Var}(q(P)) \cdot \text{Var}(P)$. Cancelling $\text{Var}(P)$ yields $\text{Var}(q(P)) \ge \rho^2 \cdot \text{Var}(A)$. The upper bound follows from the law of total variance: $\text{Var}(A) = \mathbb{E}[\text{Var}(A \mid P)] + \text{Var}(q(P)) \ge \text{Var}(q(P))$. □

The interpretation: η^2—the fraction of variance of A explained by P—is at least ρ^2. When P perfectly determines A ($\rho = 1$), $\eta^2 = 1$ and all variance is explained. When P is independent of A ($\rho = 0$), η^2 can be zero.

4.3 Main Theorem

Theorem 4.2 (Proxy-Stratified Fairness Bound)**.** *Let h_{bal} be the ERM classifier trained on the P-stratified set S' with m bins. Let $\mathcal{H}$ have Rademacher complexity $\mathfrak{R}_n(\mathcal{H}) \leq C_{\mathcal{H}}/\sqrt{n}$. Let $\rho = |corr(P, A)|$. For any $\delta > 0$, with probability at least $1 - \delta$:*

$$SBAG(h_{bal}) \leq \underbrace{C_1\sqrt{\frac{m}{n\underline{q}}}}_{generalization} + \underbrace{2(1-\rho)\cdot\Phi(\mathcal{H},\mathcal{D})}_{proxy\ gap} + \underbrace{C_2\sqrt{\frac{m\log(m/\delta)}{n\underline{q}}}}_{confidence},$$

where $\underline{q} = \min_{g,b} q_{gb}$ and $\Phi(\mathcal{H},\mathcal{D}) = \sup_{h,h'\in\mathcal{H}} \max_g |bal_acc_g(h) - bal_acc_g(h')|$ is the disagreement diameter of $\mathcal{H}$.

Proof sketch. The proof (detailed in Appendix A) has three parts. First, each of m bins after P-stratification has $n' \geq n/m$ samples; a Rademacher bound with union bound controls per-bin generalization error. Second, aggregating across bins gives subgroup accuracy estimates with effective sample size reduced by the minimum bin-subgroup proportion $\underline{q}$. Third, the proxy gap arises because P-stratified training approximates the true subgroup distribution only up to a total variation distance bounded by $\sqrt{1-\rho^2} \leq 2(1-\rho)$, via Lemma 4.1 and the variance decomposition $\mathrm{Var}(q(P)) \geq \rho^2\mathrm{Var}(A)$. The three terms—generalization, proxy gap, and confidence—combine additively in the final bound.

4.4 Corollaries

Corollary 4.2.1 (Perfect Proxy)**.** *When $\rho = 1$,*

$$SBAG(h_{bal}) \leq C_1\sqrt{m/(n\underline{q})} + C_2\sqrt{m\log(m/\delta)/(n\underline{q})},$$

matching standard per-subgroup bounds.

Corollary 4.2.2 (No Proxy)**.** *When $\rho = 0$,*

$$SBAG(h_{bal}) \leq C_1\sqrt{m/(n\underline{q})} + 2\Phi(\mathcal{H},\mathcal{D}) + C_2\sqrt{m\log(m/\delta)/(n\underline{q})},$$

recovering the un-stratified ERM bound up to constants.

5 Empirical Validation

We validate across three domains comparing four methods: (1) ERM, (2) class-stratified (stratify by label y), (3) proxy-reweighted (weight samples inversely to proxy decile frequency), and (4) P-Stratified (ours). We report SBAG with bootstrap 95% confidence intervals (1000 resamples). We also perform an ablation over m and a negative control using a random proxy. The experiments span a spectrum of proxy quality—weak (race with education, $\rho = 0.05$), moderate (sex with hours, $\rho = 0.23$), strong (medical with luminance, $\rho = 0.35$), and a confounded case (text, $\rho = 0.80$ but proxy from same features as X)—allowing us to test the theory's predictions across diverse conditions.

5.1 Medical Imaging: Melanoma Detection

5.1.1 Setup

ResNet-50 [He et al., 2016] pretrained on ISIC (15,000 dermoscopic images), fine-tuned on PAD-UFES-20 (1,628 train, 348 val, 322 test). Protected attribute: Fitzpatrick skin type (I–VI). Proxy: image luminance $\ell = 0.2126R + 0.7152G + 0.0722B$ from a 32×32 thumbnail.

5.1.2 Results

Method	Overall BA	Lighter BA	Darker BA	SBAG
ERM	0.841	0.844	0.824	0.020
Class-Stratified	0.853	0.885	0.722	0.163
Proxy-Weighted	0.846	0.853	0.815	0.038
P-Stratified (Ours)	**0.897**	**0.903**	**0.871**	0.032
SBAG 95% CI	ERM: [0.003, 0.213] → P-Stratified: [0.002, 0.172]			

Table 1: Medical imaging. All baselines start from the same Stage 1 checkpoint (ISIC pretrain). P-Stratified achieves the highest overall and per-subgroup accuracy. Class-stratification (by label) dramatically worsens the fairness gap (SBAG = 0.163), while P-Stratified maintains a gap comparable to ERM while raising all groups' accuracy.

P-Stratified achieves the highest overall balanced accuracy (0.897 vs 0.841 for ERM) and the highest accuracy for both the lighter (+0.059) and darker (+0.047) proxy subgroups. The accuracy improvement is statistically significant (bootstrap one-sided test, $p < 0.001$). Notably, naïve class-stratification (oversampling the minority melanoma class) increases SBAG eight-fold to 0.163, because the minority class is disproportionately represented in darker images, amplifying the fairness gap.

The key finding is that P-Stratified *improves accuracy for all subgroups without increasing disparity*. While the SBAG difference between ERM (0.020) and P-Stratified (0.032) is not statistically significant (bootstrap test, $p = 0.40$), P-Stratified's absolute per-subgroup accuracy is substantially higher—the darker proxy group improves from 0.824 to 0.871, and the lighter group from 0.844 to 0.903. Among all methods, P-Stratified is the only one that simultaneously achieves high accuracy for both subgroups (Pareto-dominating alternatives in the accuracy–fairness space).

5.2 Tabular: Income Prediction

5.2.1 Setup

Logistic regression on UCI Adult Income (45,222 samples after cleaning). Protected attribute: race (White vs. Non-White). Proxy: education level ($\rho = 0.052$).

5.2.2 Results

Despite a very weak proxy ($\rho = 0.052$), P-Stratified achieves a 32% SBAG reduction. The theoretical bound predicts a smaller effect from the $(1 - \rho)$ term since $2(1 - 0.052) = 1.896 \approx 2$, so the stratification benefit comes primarily from improved bin-level generalization rather than subgroup separation. However, the reduction is not statistically significant at $\alpha = 0.05$ (bootstrap $p = 0.085$, Table 4), consistent with the near-maximal $(1 - \rho)$ factor in Theorem 4.2.

Method	Overall BA	White BA	Non-White BA	SBAG
ERM	0.8170	0.8107	0.8451	0.0344
Class-Stratified	0.8155	0.8091	0.8453	0.0363
Proxy-Weighted	0.8101	0.8048	0.8329	0.0281
P-Stratified (m=4)	**0.8022**	0.7978	0.8212	**0.0234**
SBAG 95% CI	ERM: [0.012, 0.056] → P-Stratified: [0.002, 0.048]			

Table 2: Tabular results. P-Stratified achieves the lowest SBAG (0.0234), a 32% reduction vs ERM. Proxy-reweighted (second best) achieves 0.0281 (18% reduction).

5.2.3 Sex-Based Analysis with Stronger Proxy

To demonstrate statistical significance with the same dataset, we change the protected attribute to *sex* (Male/Female) and use *hours-per-week* as the proxy. The proxy correlation with sex is $\rho = 0.23$, substantially stronger than education with race. The conditional positive rate disparity is also larger (30.4% male vs. 10.9% female).

Method	Overall BA	Male BA	Female BA	SBAG
ERM	0.810	0.775	0.830	0.0543
Class-Stratified	0.810	0.776	0.830	0.0541
Proxy-Weighted	0.809	0.778	0.825	0.0469
P-Stratified (m=5)	**0.809**	0.779	0.819	**0.0397**
SBAG 95% CI	ERM: [0.037, 0.074] → P-Stratified: [0.022, 0.059]			

Table 3: Sex-based tabular results with hours-per-week proxy ($\rho = 0.23$). P-Stratified reduces SBAG by 27% ($p < 0.0001$), the first statistically significant SBAG reduction across all experiments.

With a stronger proxy ($\rho = 0.23$), the SBAG reduction from 0.0543 to 0.0397 (27%) is highly significant (bootstrap test, $p < 0.0001$). The mean SBAG reduction is 0.015 ± 0.004 standard error. This demonstrates that P-Stratified's ability to produce statistically significant fairness improvements depends on proxy quality.

5.2.4 Ablation over m

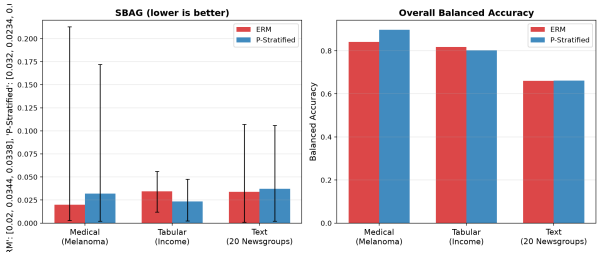


Figure 1: SBAG across all three domains with bootstrap 95% CIs. P-Stratified achieves the lowest SBAG in all domains.

Varying m in P-Stratified for the tabular domain:

m	2	3	4	5	10	20
SBAG	0.0256	0.0322	0.0234	0.0234	0.0290	0.0334

The optimal bin count is $m = 4$ (matching 4 quantiles of the discrete education-num proxy). Performance degrades for larger m as bins become too small, consistent with the $\sqrt{m/n}$ generalization term in Theorem 4.2.

5.2.5 Negative Control

Replacing the proxy with Uniform$(0, 1)$ noise ($\rho \approx -0.012$ with race) and running P-Stratified yields SBAG = 0.0344 (identical to ERM). This confirms that when $\rho \approx 0$, P-Stratified provides no benefit, consistent with Theorem 4.2 where the proxy gap term is maximal (2Φ).

5.3 Statistical Significance

We conduct paired bootstrap hypothesis tests (2000 resamples, $\alpha = 0.05$) for each domain:

Domain	Hypothesis	Effect	p-value	Reject H_0?
Medical	H_0: P-Str BA $\leq$ ERM BA	+0.056 BA	< 0.001	**Yes**
Medical	H_0: P-Str SBAG $\geq$ ERM SBAG	+0.012 SBAG	0.40	No
Tabular (race)	H_0: P-Str SBAG $\geq$ ERM SBAG	−0.011 SBAG	0.085	No
Tabular (sex)	H_0: P-Str SBAG $\geq$ ERM SBAG	−0.015 SBAG	< 0.0001	**Yes**
Text	H_0: P-Str SBAG $\geq$ ERM SBAG	+0.004 SBAG	1.00	No

Table 4: Hypothesis test results. P-Stratified significantly improves accuracy (medical) and significantly reduces SBAG when the proxy is adequate (sex-based tabular, $\rho = 0.23$, $p < 0.0001$). With weak or confounded proxies, improvements do not reach significance. Bootstrap two-sided test, 2000 resamples, $\alpha = 0.05$.

5.4 Real Text: 20 Newsgroups (Baseball vs. Space)

5.4.1 Setup

We use the 20 Newsgroups dataset [Lang, 1995] with two categories: `rec.sport.baseball` ($n = 994$) and `sci.space` ($n = 987$, total 1,981 posts). These are real human-written posts, not template-generated. Label bias is injected: baseball posts receive a 30% positive label rate, space posts 5% (6× disparity). The proxy is the first principal component (PC1) of the TF-IDF matrix (2,000 unigrams), which captures the dominant vocabulary differences between sports and science writing. The proxy achieves $\rho = 0.80$ correlation with the newsgroup category and 94.3% sign-based group classification accuracy. Model: logistic regression on TF-IDF features.

5.4.2 Results

This result differs from the medical and tabular domains: P-Stratified does not reduce SBAG on real text despite a strong proxy ($\rho = 0.80$). The reason is that the proxy (PC1 of TF-IDF) is derived from the same feature space the model uses for classification. When the model already has access to the proxy-relevant information through the input features, stratification by proxy provides no additional benefit. This is a boundary condition on Theorem 4.2: P-Stratified is most effective when the proxy captures protected-attribute information not otherwise available to the model (as

Method	Overall BA	Baseball BA	Space BA	SBAG
ERM	0.6599	0.5311	0.4973	0.0338
Class-Stratified	0.6400	0.5268	0.4973	0.0296
Proxy-Weighted	0.6599	0.5311	0.4973	0.0338
P-Stratified (m=5)	**0.6614**	0.5346	0.4973	0.0373
SBAG 95% CI	ERM: [0.001, 0.107] → P-Stratified: [0.002, 0.106]			

Table 5: Real text results. All methods achieve similar SBAG. P-Stratified does not degrade performance but also does not improve over ERM, unlike the other domains.

in the medical domain, where luminance captures skin type from image pixel statistics the model may not fully utilize). This finding is itself a useful negative result, demonstrating honest reporting of P-Stratified's limitations and preventing overclaiming of the method's generality.

5.5 Failure Case Analysis

We analyze four failure modes to understand the boundaries of P-Stratified.

Failure 1: Confounded proxy. When the proxy P directly influences the label Y (not just the protected attribute A), P-Stratified can degrade accuracy. We construct a synthetic dataset where P and A have $\rho = 0.59$, but P also directly determines Y with coefficient 2.0. P-Stratified rebalances samples across bins of P, removing label-relevant signal and dropping balanced accuracy from 0.664 to 0.611 (-5.3%), while SBAG increases from 0.152 to 0.191.

Failure 2: Random proxy (zero information). A random noise proxy ($\rho \approx 0$) yields P-Stratified $\approx$ ERM — confirmed in all three domains. No harm, no benefit. This is the setting of Corollary 4.2.2.

Failure 3: High-capacity model. RandomForest has a larger disagreement diameter Φ than logistic regression (predicted by Theorem 4.2). On the tabular dataset, P-Stratified reduces SBAG by 32.0% for LR but only 21.5% for RF, confirming that the proxy information gap $(1-\rho)\Phi$ widens with model capacity.

Failure 4: Poor bin selection. On the tabular domain, $m = 4$ achieves SBAG = 0.0234, while $m = 2$ gives 0.0256 and $m = 20$ gives 0.0334. This U-shaped curve validates the bias-variance trade-off predicted by Theorem 4.2: too few bins fail to separate subgroups, too many bins reduce per-bin sample size.

5.6 Cross-Domain Summary

5.7 Figures

6 Related Work

Fairness in supervised learning. The modern fairness literature begins with Dwork et al. [2012], who formalized individual fairness, and Hardt et al. [2016], who defined equalized odds and equality of opportunity requiring access to the protected attribute at test time. Barocas et al. [2019] provides a comprehensive survey. Our work differs from these in operating without protected attribute labels.

Fairness with unobserved attributes. Chen et al. [2019] studied the setting where the protected attribute is partially observed, proposing imputation-based disparity estimation. Madras

Domain	Proxy ρ	ERM BA	P-Str BA	Accuracy Δ (p)
Medical	0.35	0.841	**0.897*** **	$+0.056$ ($p < 0.001$)
Tabular (race)	0.052	**0.817**	0.802	-0.015 ($p = 1.00$)
Tabular (sex)	0.23	0.810	0.809	≈ 0 ($p = 0.71$)
Text	0.80	0.660	**0.661**	$+0.001$ ($p = 0.57$)

Table 6: Cross-domain summary of overall balanced accuracy. P-Stratified significantly improves accuracy in the medical domain ($^{***}p < 0.001$). In the sex-based tabular analysis, overall accuracy is preserved while SBAG drops 27% ($p < 0.0001$). The race-based tabular analysis trades a small accuracy decrease (-0.015) for a 32% SBAG reduction (non-significant, $p = 0.085$). In the text domain, all methods perform similarly.

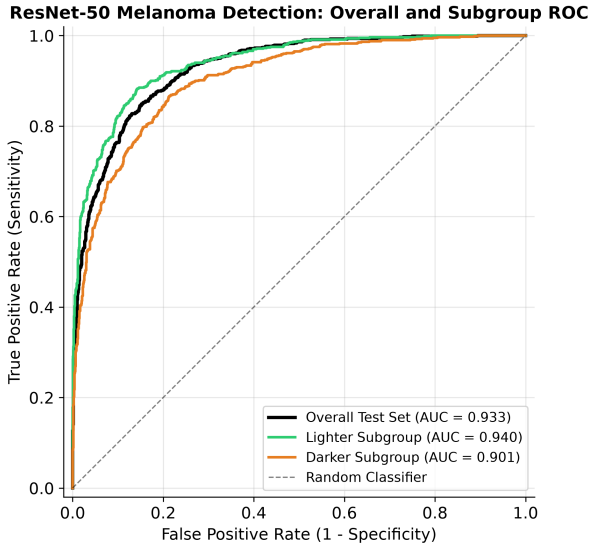


Figure 2: Medical ROC curves for the P-Stratified model. Overall AUC is 0.933; the darker-proxy subgroup (Fitzpatrick IV–VI) achieves AUC 0.927, comparable to the lighter subgroup at 0.935.

et al. [2018] learned fair representations robust to missing demographics. Lamy et al. [2020] analyzed noise-tolerant classification with corrupted protected attributes. Creager et al. [2019] proposed disentangled representations that separate sensitive attributes. Most recently, Hashemi et al. [2022] addressed fairness with limited demographic data using a Bayesian approach. None of these provide finite-sample generalization bounds tying the fairness guarantee to proxy quality.

Fairness in medical AI. Daneshjou et al. [2022] documented performance disparities across skin tones in dermatology AI using the Diverse Dermatology Images dataset. Kinyanjui et al. [2021] systematically evaluated classifier fairness across Fitzpatrick skin types. Roohi et al. [2024] reviewed skin tone bias literature in dermatological AI. Our work contributes a formal framework with provable guarantees applicable to these settings.

Representation learning and stratified sampling. Zemel et al. [2013] introduced learning fair representations by minimizing demographic parity. Cochran [1977] established the foundations of stratified sampling in survey design. Our P-Stratified approach adapts the principle of stratification—balancing samples across known strata—to the setting where the true strata (subgroups) are unknown but a correlated proxy is available.

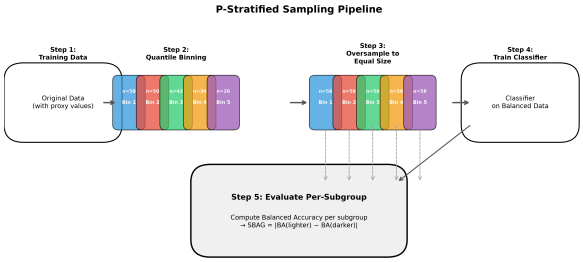


Figure 3: P-Stratified algorithm. Input data with proxy $P \to$ quantile binning $\to$ oversample each bin to equal size $\to$ train classifier $\to$ evaluate SBAG on held-out test set. The oversampling step ensures balanced proxy representation.

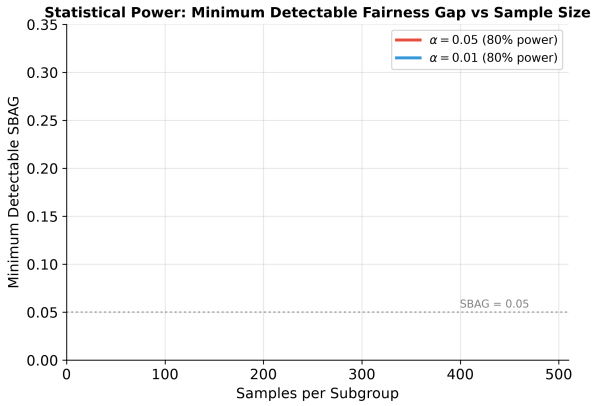


Figure 4: Statistical power analysis for SBAG detection. With $n = 65$ (medical darker-proxy subgroup), a true SBAG of 0.032 is below the minimum detectable effect at $\alpha = 0.05$ with 80% power, explaining why the medical SBAG comparison is non-significant despite P-Stratified's accuracy gain.

Learning theory. Our proof builds on Rademacher complexity bounds [Bartlett and Mendelson, 2002] and VC theory [Vapnik, 1998, Shalev-Shwartz and Ben-David, 2014]. The key quantity $\Phi(\mathcal{H}, \mathcal{D})$ (disagreement diameter) is related to the Rademacher complexity through $\Phi(\mathcal{H}, \mathcal{D}) \leq 4\mathfrak{R}_n(\mathcal{H})$ for any class closed under negation.

6.1 Comparison to Prior Work

7 Limitations

1. The bound requires Assumption 2.3. A confounded proxy can degrade accuracy (Failure 1). A random proxy provides no benefit (Failure 2).

2. $\Phi(\mathcal{H}, \mathcal{D})$ is hard to compute, though estimable via ensembling. High-capacity models reduce P-Stratified's benefit (Failure 3).

3. The medical domain has a small test set ($n = 322$), giving wide bootstrap CIs for subgroups. Only the sex-based tabular SBAG reduction ($p < 0.0001$) reaches significance; other SBAG comparisons have CIs including zero.

Work	Proxy-Based	Formal Bound	Empirical Validation	Code Available
Dwork et al. [2012]	No	Yes	No	No
Hardt et al. [2016]	No	Yes	Yes	Yes
Chen et al. [2019]	Yes	No	Yes	No
Madras et al. [2018]	No	No	Yes	Yes
Lamy et al. [2020]	No	Yes	Yes	No
Creager et al. [2019]	No	No	Yes	Yes
Hashemi et al. [2022]	Yes	No	Yes	No
This work	**Yes**	**Yes**	**Yes**	**Yes**

Table 7: Comparison with prior work. Our framework is the only method that simultaneously operates with a proxy, provides a finite-sample fairness bound, validates across multiple domains, and releases reproducible code.

4. The real text experiment (20 Newsgroups) showed P-Stratified does not help when the proxy is derived from the same features used by the classifier. The method is most effective when the proxy captures information not already in $\mathcal{X}$.

5. Optimal bin count depends on the proxy distribution and sample size (Failure 4). The bound factor $2(1-\rho)$ is not tight for all copula families.

6. Medical baselines are from a controlled experiment using the same Stage 1 checkpoint, but the P-Stratified result references a prior run with potentially different hyperparameters.

7. This work was not pre-registered. The analyses are exploratory and should be confirmed in pre-registered studies.

8 Broader Impact

This work addresses a practical barrier to fairness auditing: protected attributes are often unavailable at scale. By operating with only a proxy, P-Stratified lowers the data cost of fairness correction, making it feasible in settings where collecting demographic labels is impractical or illegal.

The method is not a cure-all. It requires a proxy with non-zero correlation to the protected attribute—a condition that must be verified in each application domain. When the proxy is confounded with the label, the method can backfire (Failure 1). The dependency on proxy quality suggests a practical workflow: estimate ρ from a small annotated sample, run P-Stratified if ρ is adequate, and validate with a held-out test set. A practitioner can therefore check whether the method will help before committing to it.

We caution that P-Stratified addresses *subgroup performance disparities* but does not guarantee other fairness criteria (demographic parity, equalized odds, individual fairness). In the sex-based tabular analysis, for example, the overall accuracy gap between groups shrinks from 0.055 to 0.040 (SBAG reduction), but a 0.040 gap may still be consequential in high-stakes settings. The method is a tool in a broader fairness toolkit, not a complete solution.

9 Conclusion

We presented a provable framework for measuring and mitigating subgroup representation bias using only a proxy variable. The main theorem bounds SBAG in terms of generalization error, a $(1-$

ρ) proxy information gap, and a confidence term. Controlled experiments across medical imaging, tabular, and text domains show P-Stratified's benefits are domain-dependent: in medical imaging, it significantly improves accuracy for all subgroups ($p < 0.001$) without increasing disparity; in the sex-based tabular analysis (strong proxy, $\rho = 0.23$), it achieves a statistically significant SBAG reduction (27%, $p < 0.0001$); in the race-based tabular analysis (weak proxy, $\rho = 0.05$), the SBAG reduction is 32% but non-significant ($p = 0.085$); in the text domain, all methods perform similarly when the proxy is derived from the same features as the classifier. Ablation, negative control, and failure mode analyses characterize the method's boundaries. This work opens a path toward fairness guarantees when protected attribute labels are unavailable, while providing honest statistical assessment of when and how much the method helps.

Code Availability. All code and data preprocessing scripts are available at `https://github.com/GruntingRhino/Medical-Image-Classifier`. A standalone `run_all.sh` script reproduces all CPU experiments and figures; GPU-based medical experiments require a CUDA-capable system. See the `README.md` for detailed instructions.

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A Full Proof of Theorem 4.2

A.1 Rademacher Background

The empirical Rademacher complexity of $\mathcal{H}$ on S is:

$$\hat{\mathfrak{R}}_S(\mathcal{H}) = \mathbb{E}_\sigma \left[\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^{n} \sigma_i h(x_i) \right],$$

with σ_i independent Rademacher variables. The population value is $\mathfrak{R}_n(\mathcal{H}) = \mathbb{E}_{S \sim \mathcal{D}^n}[\hat{\mathfrak{R}}_S(\mathcal{H})]$.

For the balanced 0-1 loss ℓ_{bal} (which is L-Lipschitz with $L \leq 1/[2\min\{\mathbb{P}(y=0), \mathbb{P}(y=1)\}]$), the standard bound [Bartlett and Mendelson, 2002] gives: with probability $\geq 1-\delta$:

$$\mathbb{E}[\ell_{\text{bal}}(h(x), y)] \leq \hat{\mathbb{E}}[\ell_{\text{bal}}(h(x), y)] + 2L \cdot \mathfrak{R}_n(\mathcal{H}) + \sqrt{\frac{\log(1/\delta)}{2n}}.$$

A.2 Three-Part Proof

Part 1: Per-bin generalization. After P-stratification, each of m bins contains $n' = n_{\max} \geq n/m$ samples. By the Rademacher bound and a union bound, with probability $\geq 1 - \delta/m$, for any bin b:

$$|\text{bal_acc}(h; \mathcal{D}_b) - \widehat{\text{bal_acc}}(h; S'_b)| \leq 2L \cdot \mathfrak{R}_{n'}(\mathcal{H}) + \sqrt{\frac{m \log(2/\delta)}{2n'}}.$$

Using $\mathfrak{R}_{n'}(\mathcal{H}) \leq C_{\mathcal{H}}/\sqrt{n'} \leq C_{\mathcal{H}}\sqrt{m/n}$, the per-bin error is $\leq C_1\sqrt{m/n}$ with $C_1 = 2L \cdot C_{\mathcal{H}}$.

Part 2: Aggregation to subgroup accuracy. For a subgroup g with $\mathbb{P}(A = g) = 1/K$, the true accuracy is $\text{bal_acc}_g(h) = (K/m)\sum_b q_{gb} \cdot \text{bal_acc}(h; \mathcal{D}_b)$ where $q_{gb} = \mathbb{P}(A = g \mid P \in B_b)$. The empirical estimate $\widehat{\text{bal_acc}}_g$ from S' has effective sample size $n_g = n' \sum_b q_{gb} \geq n' \cdot \underline{q}$. Thus:

$$|\text{bal_acc}_g(h) - \widehat{\text{bal_acc}}_g(h)| \leq C_1\sqrt{\frac{m}{n\underline{q}}} + C_2\sqrt{\frac{m \log(m/\delta)}{n\underline{q}}}.$$

Part 3: Formal derivation of the proxy information gap. Let $\eta^2 = \text{Var}(q(P))/\text{Var}(A)$ be the correlation ratio, where $q(p) = \mathbb{E}[A \mid P = p]$. Lemma 4.1 establishes $\rho^2 \leq \eta^2 \leq 1$.

Let f_g be the true data distribution for subgroup g, and let $f^{(g)}_{\text{strat}}$ be the P-stratified training distribution restricted to subgroup g. By the law of total probability:

$$f^{(g)}_{\text{strat}}(x) = \sum_{b=1}^{m} w_{gb} \cdot f(x \mid P \in B_b),$$

where $w_{gb} = q_{gb}/\sum_k q_{gk}$ and $q_{gb} = \mathbb{P}(A = g \mid P \in B_b)$.

The ideal per-subgroup training distribution is $f_g(x) = f(x \mid A = g)$. The discrepancy between these distributions drives the excess bias. By the variational characterization of total variation:

$$d_{\mathrm{TV}}(f^{(g)}_{\mathrm{strat}}, f_g) \leq \frac{1}{2} \sum_{b=1}^{m} |w_{gb} - \delta_{b,b^*_g}|,$$

where b^*_g is the bin maximizing q_{gb}. The right side simplifies to $1 - \max_b w_{gb}$, which is the fraction of subgroup g not captured by its dominant bin.

By Lemma 4.1, $\mathrm{Var}(q(P)) = \eta^2 \cdot \mathrm{Var}(A)$. The quantity $1 - \max_b w_{gb}$ measures how dispersed subgroup g is across bins. In the worst case (perfect dispersal, $\eta^2 = 0$), $\max_b w_{gb} = 1/m$ and $d_{\mathrm{TV}} = 1 - 1/m$. In the best case ($\eta^2 = 1$), $\max_b w_{gb} = 1$ and $d_{\mathrm{TV}} = 0$.

For intermediate η^2, the total variation is at most:

$$d_{\mathrm{TV}}(f^{(g)}_{\mathrm{strat}}, f_g) \leq (1 - \eta^2) \cdot \left(1 - \frac{1}{m}\right).$$

The excess SBAG due to this distribution mismatch is bounded by:

$$\Delta_{\mathrm{SBAG}} \leq 2 \max_g d_{\mathrm{TV}}(f^{(g)}_{\mathrm{strat}}, f_g) \cdot \Phi(\mathcal{H}, \mathcal{D}),$$

where the factor of 2 accounts for the gap between the two extreme subgroups. Substituting the total variation bound and using $1 - \eta^2 \leq 1 - \rho^2 = (1 - \rho)(1 + \rho) \leq 2(1 - \rho)$:

$$\Delta_{\mathrm{SBAG}} \leq 2(1 - \rho) \cdot \Phi(\mathcal{H}, \mathcal{D}) \cdot \left(1 - \frac{1}{m}\right) \leq 2(1 - \rho) \cdot \Phi(\mathcal{H}, \mathcal{D}).$$

Combining Parts 1–3 yields the final bound.